

Lab 6: Analog and Digital Conversion

- Review
 - Analog to digital conversion (see Rizzoni [2], page 718-)
 - * Analog - “... an analog signal is an electrical signal whose value varies in analogy with a physical quantity (e.g., temperature, force, or acceleration) [2].”
 - * Digital - “A digital signal ... can take on only a finite number of values [2].”
 - * Quantization - “The process of quantization consists of subdividing the range of the signal into a finite number of intervals; usually, one employs $2^n - 1$ intervals, where n is the number of bits available for the corresponding binary word [2].”
 - * Sampling - “When measurements are made of a time-varying signal (measurand) using a computerized data-acquisition system, measurements are only made at a discrete set of times, not continuously [3].”
 - * Example Circuits
 - Op amp comparator
 - tracking ADC
 - flash ADC (2 bit example)
 - other
 - Digital to analog conversion
 - * 2 bit (example)
- Equipment
 - Breadboard with triple power supply
 - Sweep/function generator (link)
 - Additional (old) function generator
 - Oscilloscope (link)
 - 741 op amp (link)
 - * $V_S^+ = 10\text{ V}$
 - * $V_S^- = -15\text{ V}$
 - Resistors: $1\text{k}\Omega$, $5\text{k}\Omega$
 - LED
 - Capacitors: $0.1\text{ }\mu\text{F}$
 - 8 switches
 - DAC 0808 (link)

* pin 1: No connection	* pin 6: A2	bit)
* pin 2: ground (GND)	* pin 7: A3	* pin 13: V_{CC}
* pin 3: V_{EE}	* pin 8: A4	* pin 14: V_{REF}^+
* pin 4: I_0	* pin 9: A5	* pin 15: V_{REF}^-
* pin 5: A1 (most sig. bit)	* pin 10: A6	* pin 16: compensa- tion
	* pin 11: A7	
	* pin 12: A8 (least sig.	

– ADC 0808 (link)

* pin 1: $v_{in,3}$	* pin 11: V_{CC}	* pin 21: 2^{-1} (most sig. bit)
* pin 2: $v_{in,4}$	* pin 12: V_{REF}^+	
* pin 3: $v_{in,5}$	* pin 13: ground (GND)	* pin 22: address lock enable (ALE)
* pin 4: $v_{in,6}$	* pin 14: 2^{-7}	* pin 23: address C
* pin 5: $v_{in,7}$	* pin 15: 2^{-6}	* pin 24: address B
* pin 6: start	* pin 16: V_{REF}^-	* pin 25: address A
* pin 7: end of conver- sion (EOC)	* pin 17: 2^{-8} (least sig. bit)	* pin 26: $v_{in,0}$
* pin 8: 2^{-5}	* pin 18: 2^{-4}	* pin 27: $v_{in,1}$
* pin 9: output enable (OE)	* pin 19: 2^{-3}	* pin 28: $v_{in,2}$
* pin 10: clock	* pin 20: 2^{-2}	

- Imagine that you just received free samples of DAC0808 and ADC0808 for you to evaluate for use in a future product. Unfortunately, no one at your current place of employment has experience using chips to convert between analog and digital signals. **Overall, your objective is to demonstrate the conversion between analog and digital signals using the DAC0808 and ADC0808 chips.**

– Digital to analog converter

- * Circuit - see page 4 of DAC0808 (link), typical application

· pin 2: to ground	· pin 6: A2: 0 or 5V
· pin 3: to -15 V	· pin 7: A3: 0 or 5V
· pin 4: to V_{in}^- on op amp (see page 4 of DAC0808 data sheet for op amp wiring)	· pin 8: A4: ground
· pin 5: A1 (most sig. bit): 0 or 5V	· pin 9: A5: ground
	· pin 10: A6: ground
	· pin 11: A7: ground
	· pin 12: A8 (least sig. bit):

- ground
- pin 13: to 5 V
- pin 14: to 10 V through $5K\Omega$ resistor
- pin 15: to ground through $5K\Omega$ resistor
- pin 16: to -15 V through $0.1 \mu F$ capacitor

In contrast to the typical application, we are not using precision resistors or voltage sources.

- * Measure the output for each high/low combination for the most significant bits.

Table 1: Task 1

A1	A2	A3	V_0	V_0 (theory)
low	low	low		
low	low	high		
low	high	low		
low	high	high		
high	low	low		
high	low	high		
high	high	low		
high	high	high		

- * Quantify the difference between the expected output and the measured output. Can the difference be explained totally by measurement uncertainty? Is the lack of precision resistors and voltage source significant?

* **Do not disassemble**

– Analog to digital converter

- * Circuit - see page 13 of ADC0808 (link), typical application

- pin 6: start: connect to pins 7 and 22
- pin 7: EOC: connect to pins 6 and 22
- pin 9: connect to 5V
- pin 10: clock: 100 kHz TTL 5V (or use the 555 oscillator)
- pin 11: connect to 5V
- pin 12: connect to 5V
- pin 13: GND: ground
- pin 16: V_{REF}^- : connect to ground
- pin 19: 2^{-3} : to ground through LED and $1K\Omega$ resistor in series
- pin 20: 2^{-2} : to ground through LED and $1K\Omega$ resistor in series
- pin 21: 2^{-1} : to ground through LED and $1K\Omega$ resistor in series
- pin 22: ALE: connect to pins 6 and 7
- pin 23: C: ground

- pin 24: B: ground
- pin 25: A: ground
- pin 26: $v_{in,0}$: 10 Hz sine wave, 4 V_{pp} , mean 2 V (old function generator)

In contrast to the typical application, we are not using precision voltage sources.

- * Determine the blinking pattern of the LED lights. (You could take a slow motion video on your phone.) What is the expected blinking pattern? Does the expected and observed blinking patterns match?

* **Do not disassemble**

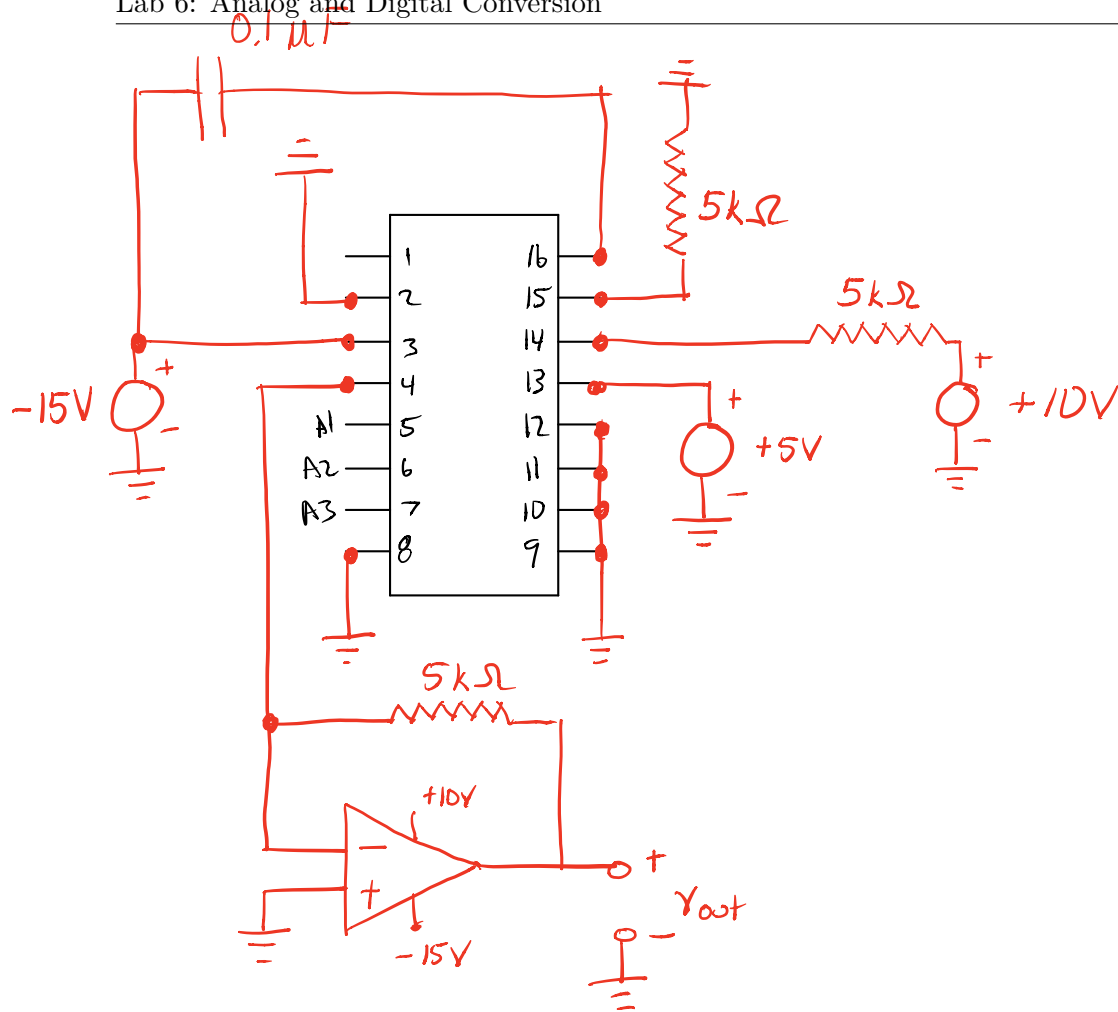
- Analog to digital and digital to analog
 - * Replace the LED/resistors with a connection to the A1, A2, and A3 pins on the DAC0808
 - * Change the input to the ADC to a 20 Hz, 4 V_{pp} sine wave with mean 2V
 - * How does the output of the DAC compare to the input of the ADC on the o-scope? Can the error be explained completely by measurement error?
- Simple ADC and DAC circuits using an op amp
 - * 741 Op amp comparator
 - v_{in}^+ to input
 - v_{in}^- to reference
 - * 741 Op amp 2 bit flash ADC (see page 719 Rizzoni)
 - use LED to read result
 - use 5V as reference voltage
 - use low frequency input in the range of 0 to 5 V as the unknown voltage
 - What is the blinking pattern of the LEDs? Does it make sense?
 - * 741 Op amp 2 bit DAC (see page 715 Rizzoni)
 - use 5V as reference voltage
 - 10 k Ω
 - 5 k Ω
 - $R_F = 2\text{k}\Omega$ (for 0 to 3 volt range)

• Lab report:

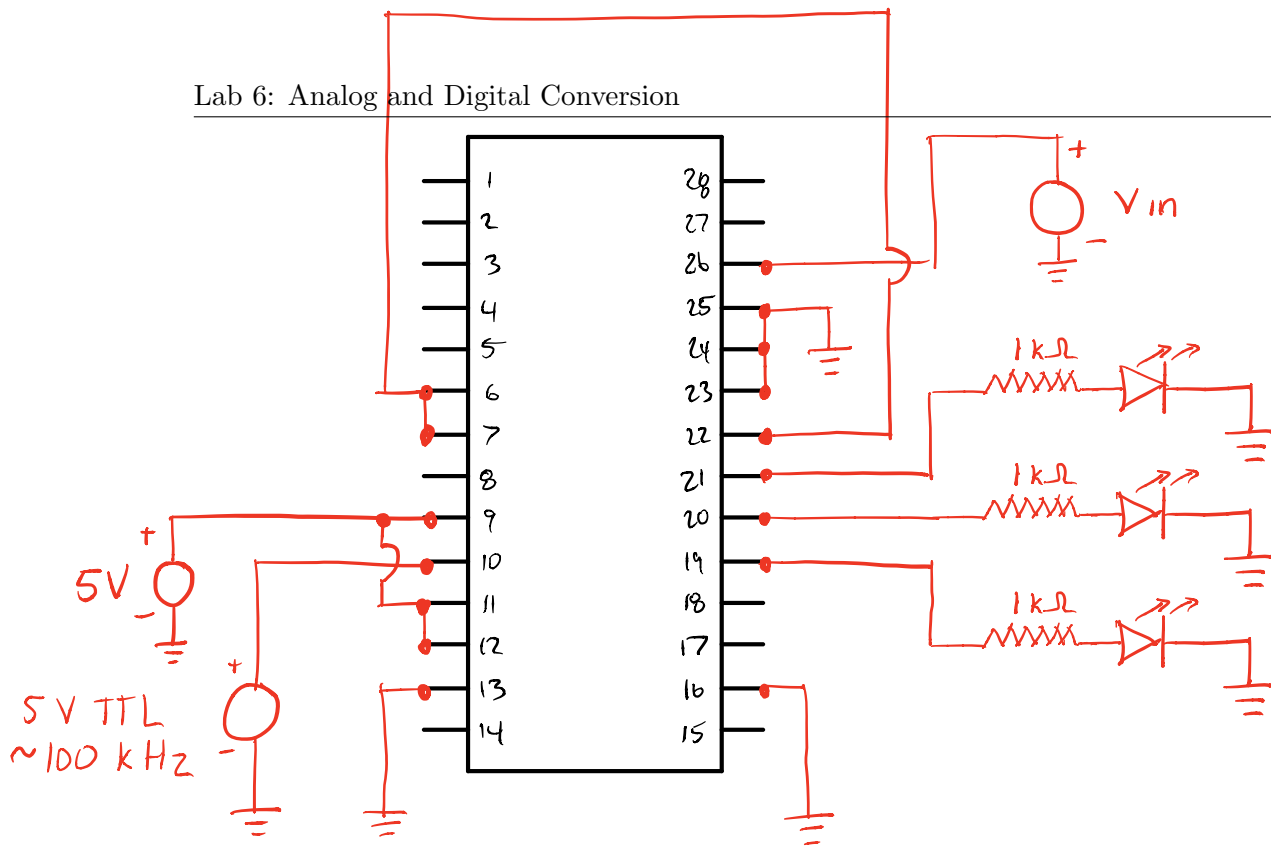
- Your lab report should follow the required lab report structure for this class.
- **Optional:** In the body of your report you should make a strong case for your objectives. Overall, your objective is to demonstrate the conversion between analog and digital signals using the DAC0808 and ADC0808 chips. Consequently, you should:

1. provide a clear description of your use of both DAC0808 and ADC0808 such that a reader could reproduce your circuits/results.
 2. discuss possible real life applications for both DAC0808 and ADC0808. Also, discuss important factors an engineer should consider when designing a system with ADC and DAC chips.
- Include the following in the Appendix:
- * This lab outline (required)
 - * Data you collected during lab (required)
 - * Discuss the difference between the measured and expected outputs for each circuit. Can you detect errors in the ADC0808 and DAC0808 chips? (optional)
 - * Any other details (data, derivations, calculations, experimental procedure, screen captures etc.)

(The purpose of the Appendix in these reports is to: (i) serve as somewhat of a lab notebook, (ii) demonstrate to the grader what you did during lab, and (iii) hold material that didn't fit into the body of the report but would be necessary to repeat the experiments or justify the claims in the report.)

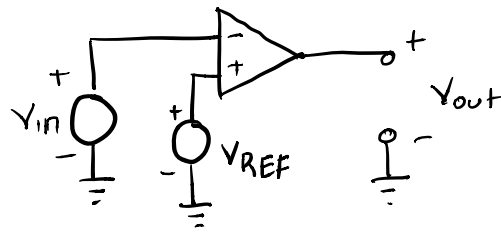


Lab 6: Analog and Digital Conversion



Comparator

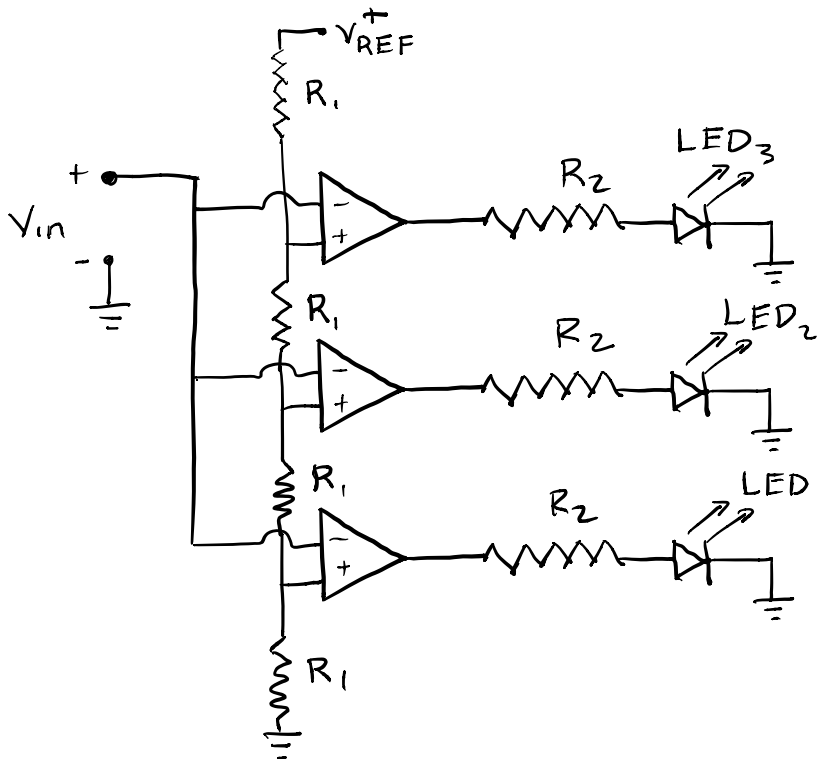
Lab 6: Analog and Digital Conversion



	V_{out}
$V_{in} < V_{REF}$	V_S^+
$V_{in} > V_{REF}$	V_S^-

2-bit flash ADC

[see Rizzoni, pp 719]

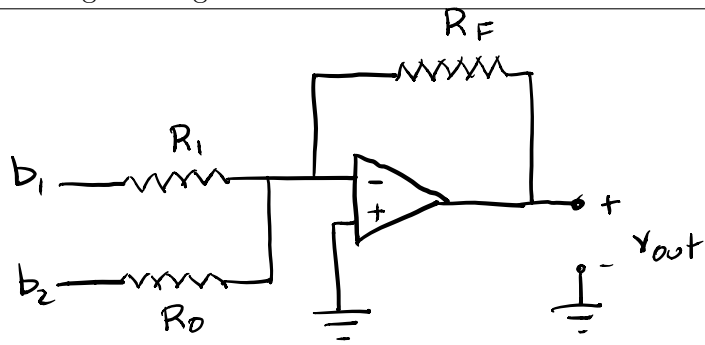


Off	On
$V_{in} > \frac{3}{4} V_{REF}$	$V_{in} < \frac{3}{4} V_{REF}$
$V_{in} > \frac{2}{4} V_{REF}$	$V_{in} < \frac{2}{4} V_{REF}$
$V_{in} > \frac{1}{4} V_{REF}$	$V_{in} < \frac{1}{4} V_{REF}$

LED ₃	LED ₂	LED ₁	
on	on	on	$V_{in} < \frac{1}{4} V_{REF}$
on	on	off	$\frac{1}{4} V_{REF} < V_{in} < \frac{1}{2} V_{REF}$
on	off	off	$\frac{1}{2} V_{REF} < V_{in} < \frac{3}{4} V_{REF}$
off	off	off	$\frac{3}{4} V_{REF} < V_{in}$

2 bit DAC

Lab 6: Analog and Digital Conversion



$$R_0 = 10 \text{ k}\Omega$$

$$R_1 = 5 \text{ k}\Omega$$

$$R_F = 2 \text{ k}\Omega$$

b_1	b_0	V_{out}
0	0	0
0	5	1
5	0	2
5	5	3